

上海科技大学
ShanghaiTech University

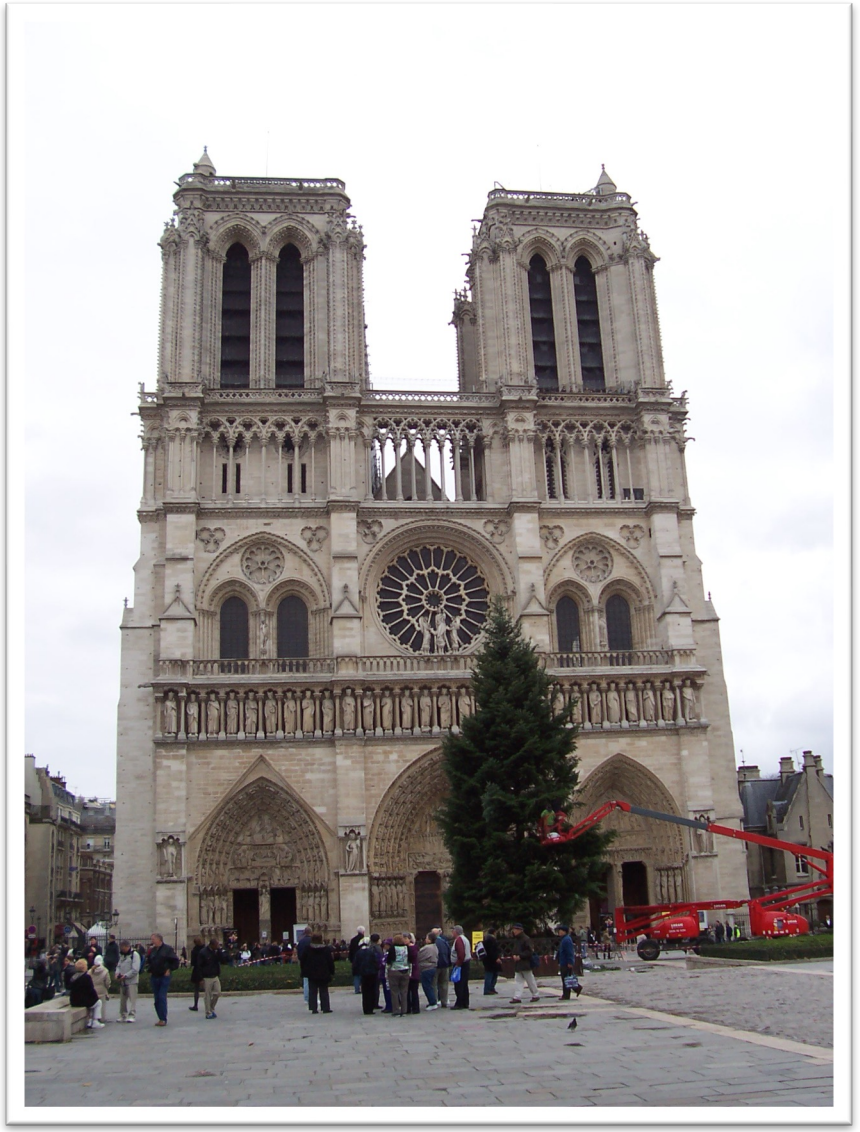
Corner Detection

CS172 Computer Vision I

Instructor: Jiayuan Gu

Invariant Feature Cross Views

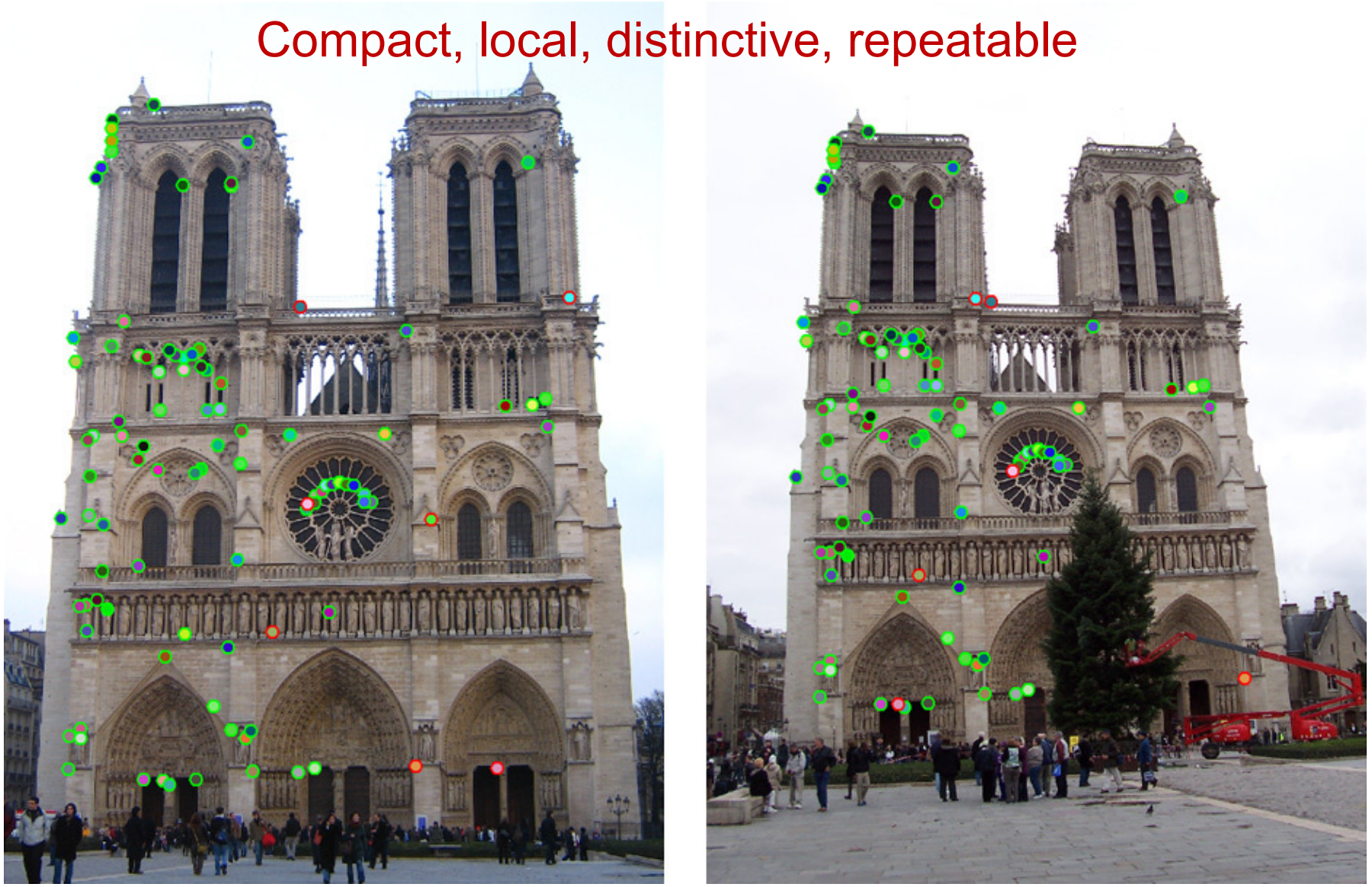




Snavely, Noah. “Photo Tourism : Exploring image collections in 3D.” *ACM Transactions on Graphics* (2006)

Interest Points or Keypoints

Compact, local, distinctive, repeatable



https://sites.cc.gatech.edu/classes/AY2016/cs4476_fall/proj2/

What are good keypoints?

- Compactness and efficiency
  - Many fewer keypoints than image pixels
- Saliency
  - Each keypoint is distinctive
- Locality
  - A keypoint occupies a relatively small area of the image; robust to clutter and occlusion
- Repeatability
  - The same keypoint can be found in several images despite geometric and photometric transformations

Source: L. Lazebnik

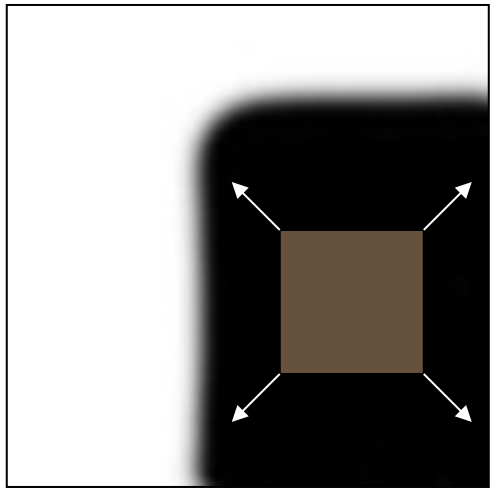
Agenda

- Roadmap
  - Corners and second-moment matrices
  - Harris detector

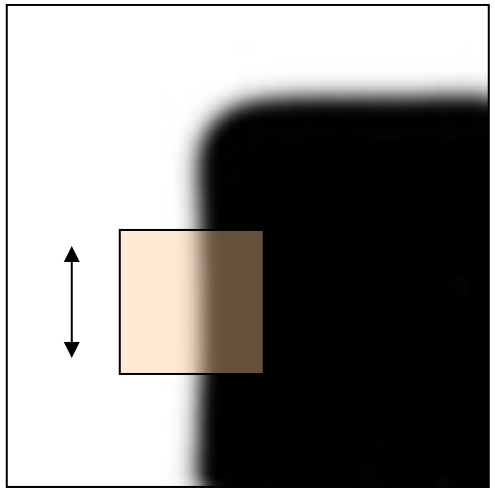
- Objectives
  - explain why corners are good keypoints
  - Derive and interpret the second-moment matrix and its eigenvalues
  - Implement the Harris corner detector
  - Identify its limitations and motivate more robust local features

Basic Idea: Detecting Corners given a Window

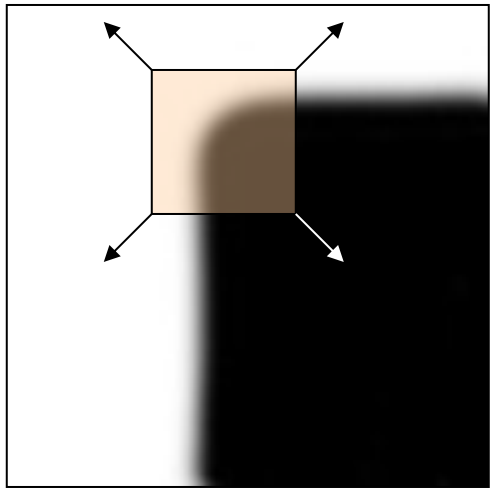
- Recognize the point by looking through a small window
- Shifting a window in *any direction* should give *a large change* in intensity



"flat" region:
no change in
all directions



"edge":
no change
along the edge
direction

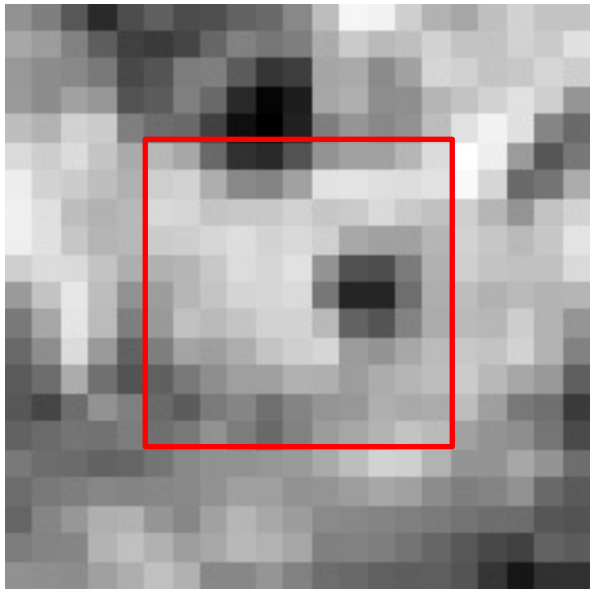


"corner":
significant
change in all
directions

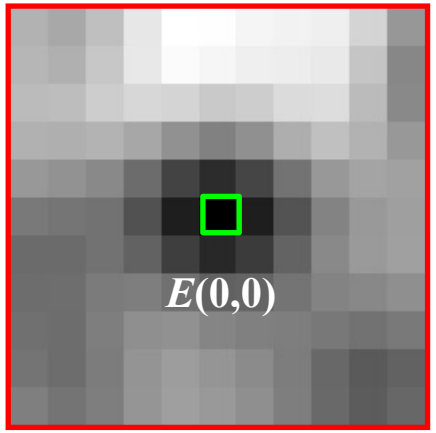
Source: L. Lazebnik

Quantifying Patch Change

$$E(u,v) = \sum_{x,y \in W} [I(x+u, y+v) - I(x,y)]^2$$



$I(x, y)$



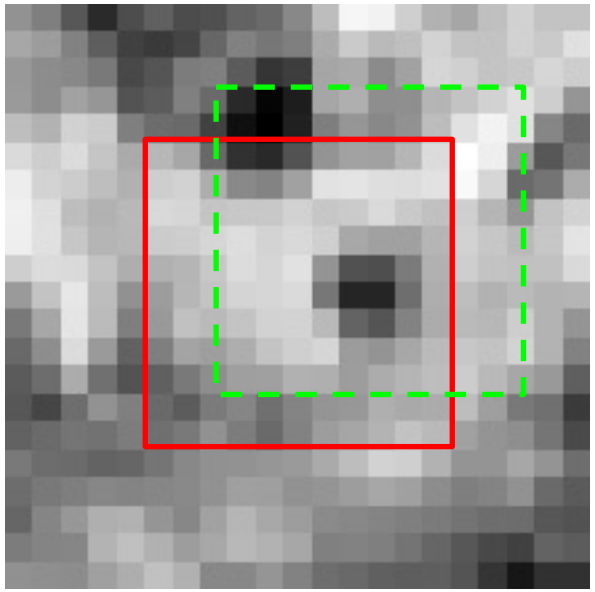
$E(u,v)$

W: neighborhood
(u,v): shifting direction of the window
$E(u,v)$: squared difference describing how much the local appearance changes

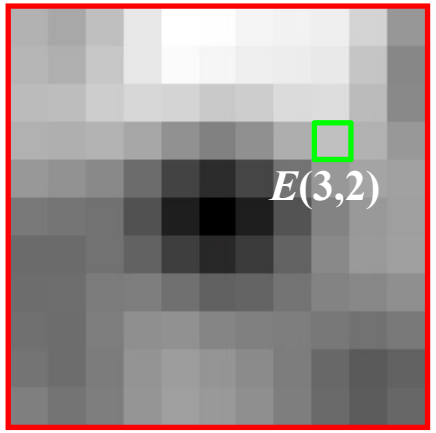
Credit: L. Lazebnik

Quantifying Patch Change

$$E(u,v) = \sum_{x,y \in W} [I(x+u, y+v) - I(x,y)]^2$$



$I(x, y)$



$E(u,v)$

W: neighborhood
(u,v): shifting direction of the window
$E(u,v)$: squared difference describing how much the local appearance changes

What shape should *E(u,v)* have near a good corner?

Credit: L. Lazebnik

First-order Taylor Approximation

- Taylor expansion at small u, v

$$I(x+u, y+v) \approx I(x,y) + I_x u + I_y v$$

  - Note that $I_x = I_x(x,y)$, The subscript is the gradient in the x-axis

- Plug into $E(u,v)$

$$E(u,v) = \sum_{x,y \in W} [I(x+u, y+v) - I(x,y)]^2$$

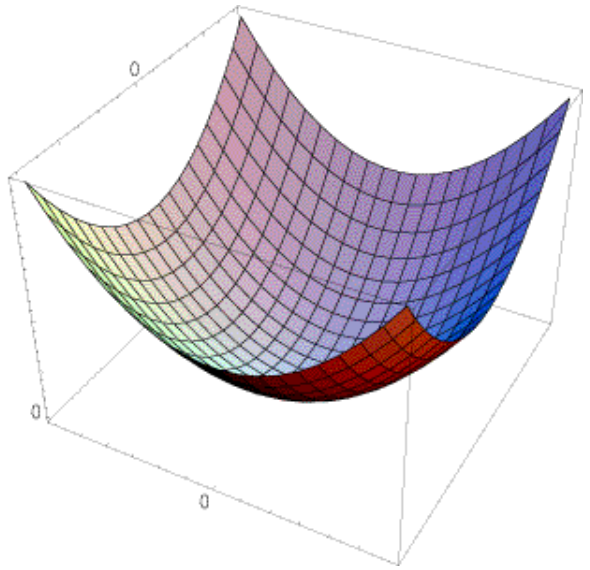
$$= \sum_{x,y \in W} \left[I_x u + I_y v\right]^2 = \sum_{x,y \in W} I_x^2 u^2 + I_y^2 v^2 + 2 I_x I_y uv$$

Credit: L. Lazebnik

Locally Approximated by Quadratic Surface

$E(u,v)$ can be locally approximated by a quadratic surface

$$E(u,v) \approx u^2 \sum_{x,y \in W} I_x^2 + v^2 \sum_{x,y \in W} I_y^2 + 2uv \sum_{x,y \in W} I_x I_y$$



In which directions does this surface have the fastest/slowest change?

Credit: L. Lazebnik

Second Moment Matrix

$E(u,v)$ can be locally approximated by a quadratic surface

$$E(u,v) \approx u^2 \sum_{x,y \in W} I_x^2 + v^2 \sum_{x,y \in W} I_y^2 + 2uv \sum_{x,y \in W} I_x I_y$$

$$= [u,v] \begin{bmatrix} \sum_{x,y \in W} I_x^2 & \sum_{x,y \in W} I_x I_y \\ \sum_{x,y \in W} I_x I_y & \sum_{x,y \in W} I_y^2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} \stackrel{\text{def}}{=} [u,v] M \begin{bmatrix} u \\ v \end{bmatrix}$$

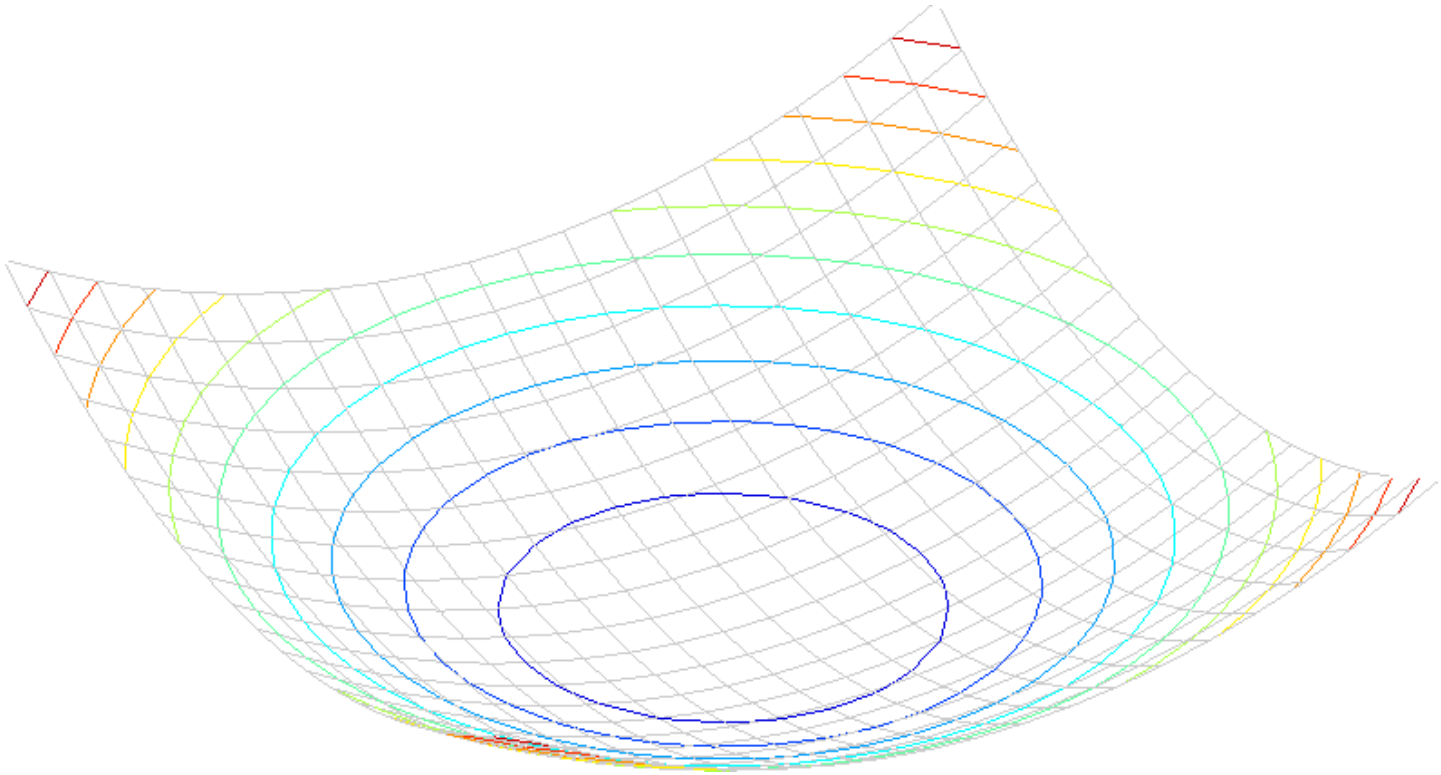
M is the second moment matrix

Credit: L. Lazebnik

Interpreting the Second Moment Matrix

A ”slice” of $E(u,v)$ is given by the equation of an ellipse

$$[u, v]M\begin{bmatrix} u \\ v \end{bmatrix} = constant$$

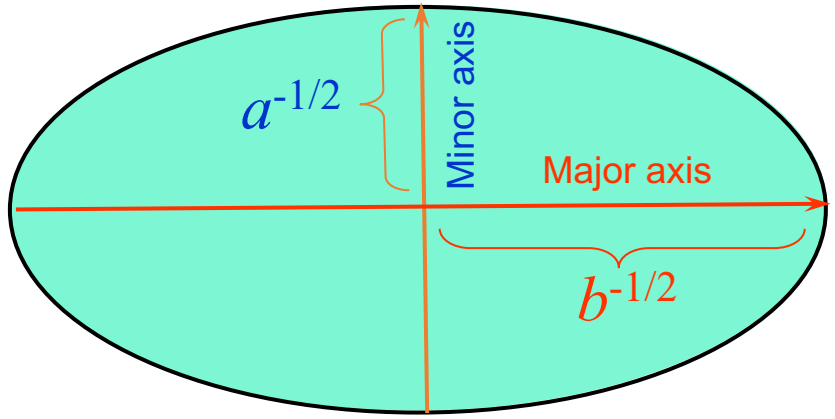


Credit: L. Lazebnik

Interpreting the Second Moment Matrix

Consider the axis-aligned case
(gradients are either horizontal or vertical)

$$[u, v] M \begin{bmatrix} u \\ v \end{bmatrix} = [u, v] \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 1$$



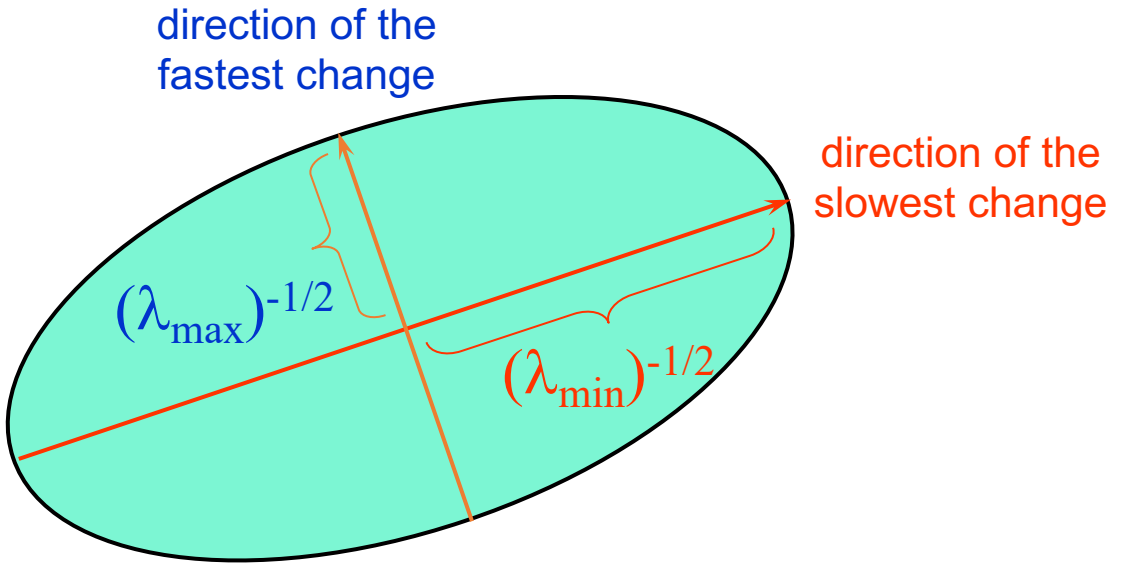
flat: a and b both close to 0
edge: a or b is close to 0
corner: neither a nor b is close to 0

Credit: L. Lazebnik

Interpreting the Second Moment Matrix

In the general case, we use eigen-decomposition of M

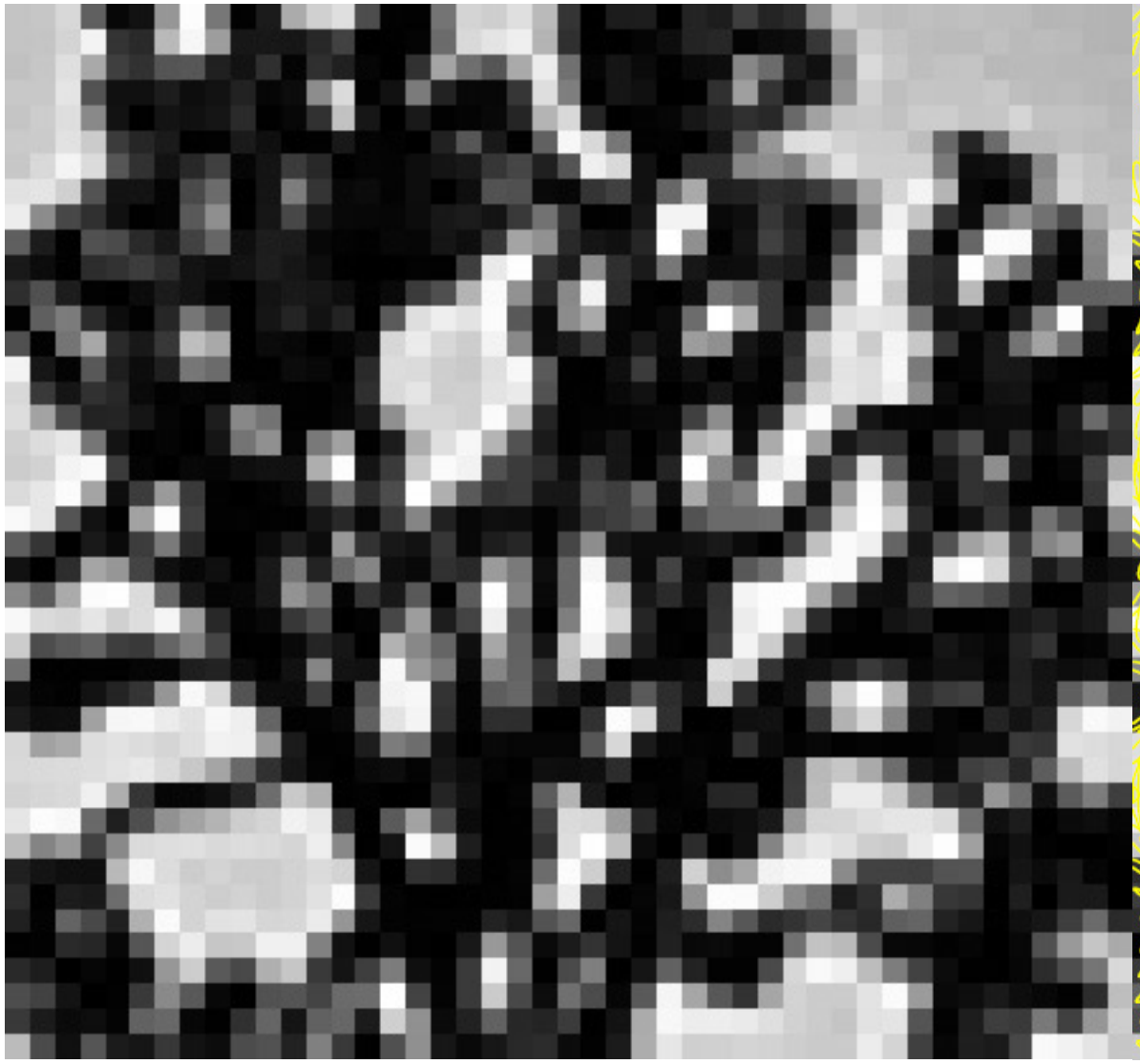
$$[u, v]M\begin{bmatrix} u \\ v \end{bmatrix} = [u, v]R^{-1}\begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}R\begin{bmatrix} u \\ v \end{bmatrix} = 1$$



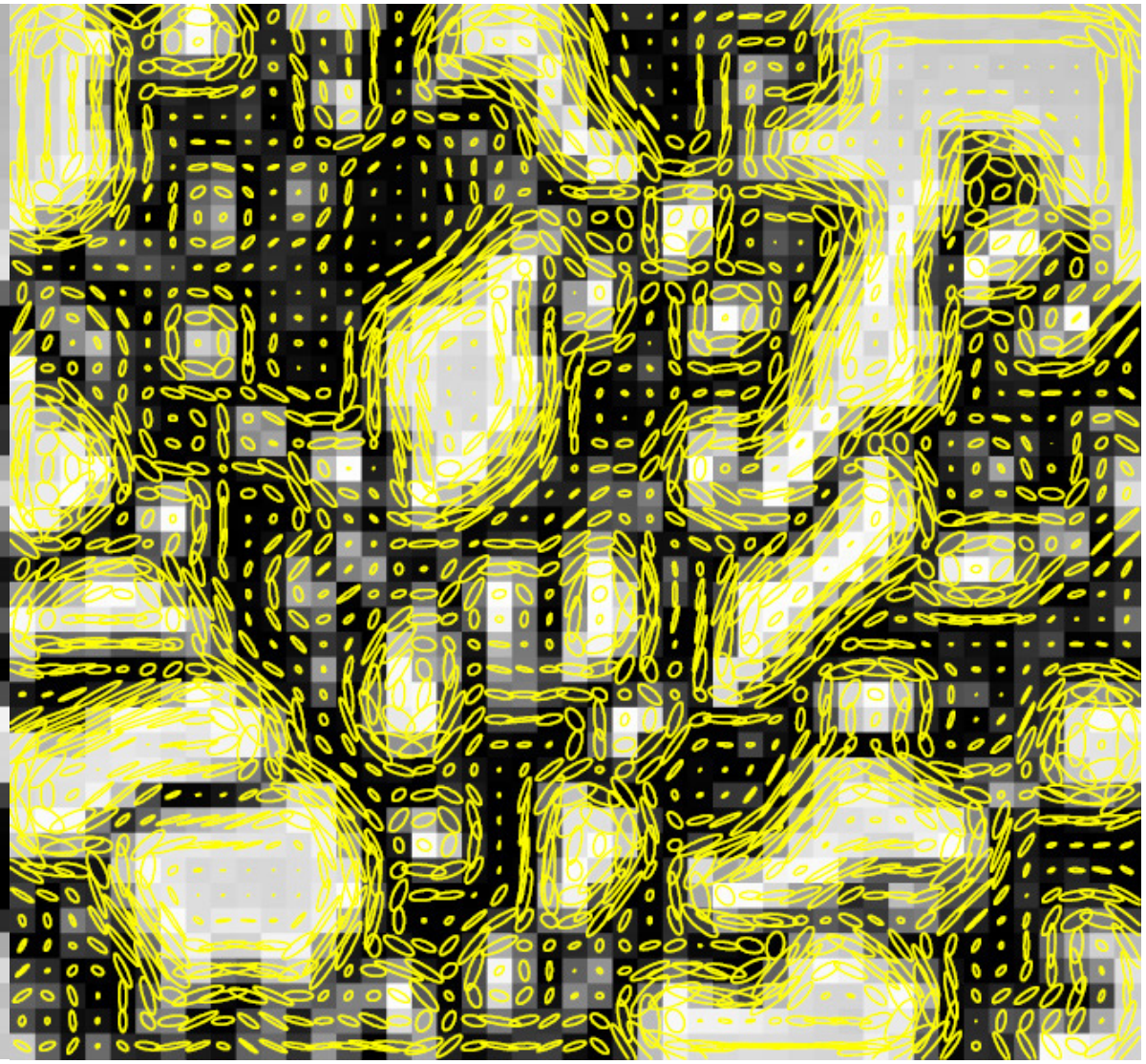
Thus M summarizes the **distribution of gradient directions and strengths in the neighborhood**

Credit: L. Lazebnik

Visualization of the Second Moment Matrix



Input image



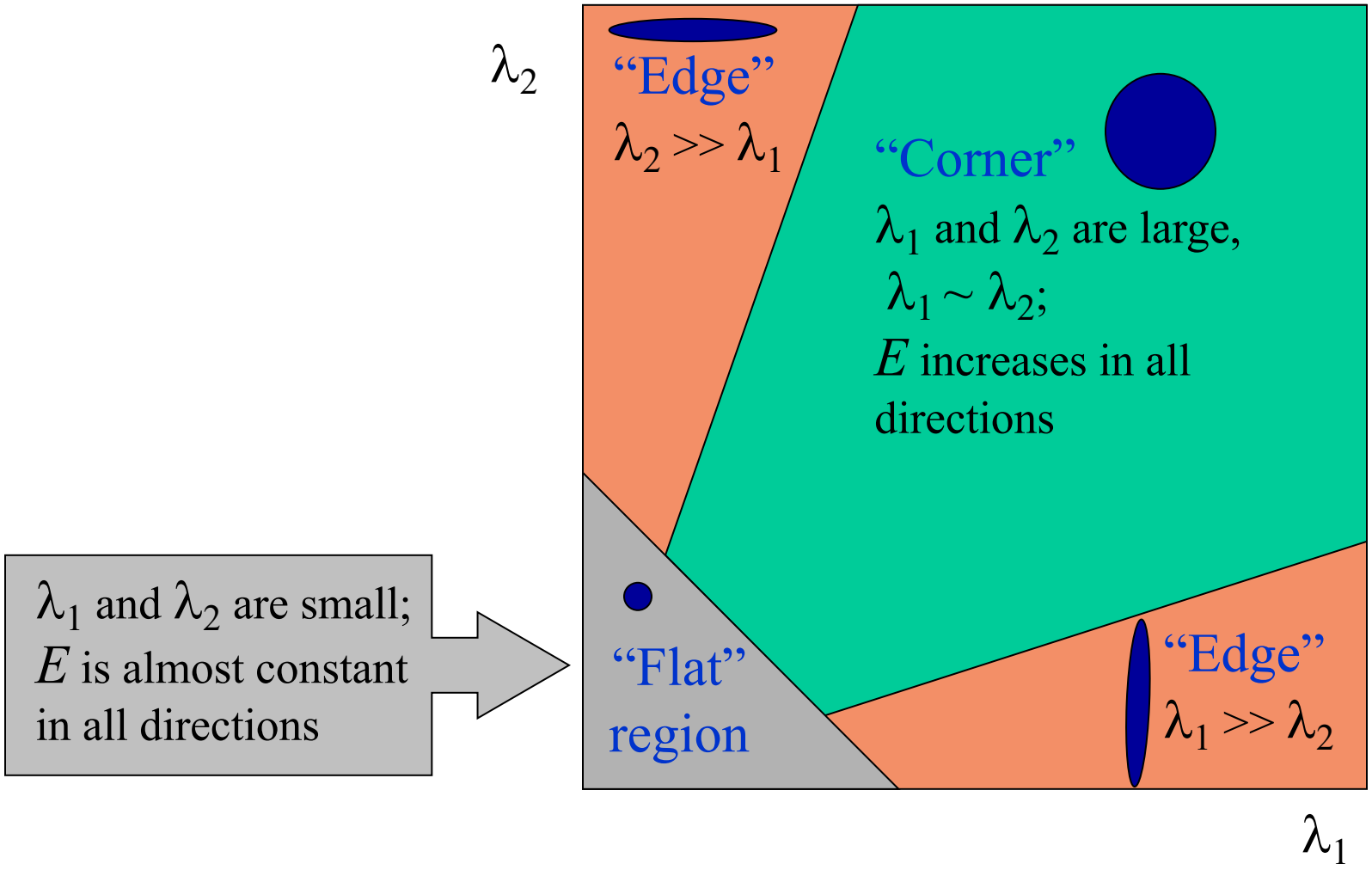
Visualization

Credit: L. Lazebnik

Harris Corner Detector

Detecting Corners by Eigenvalues

- Classification of corners using eigenvalues of *M*

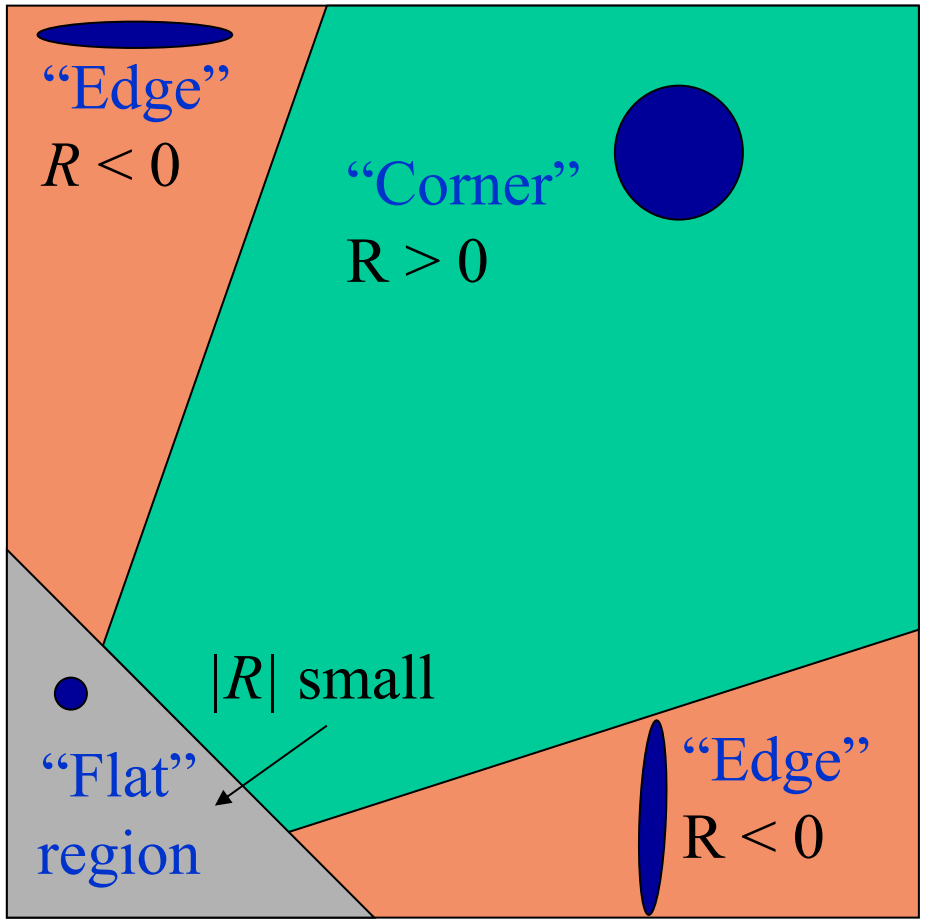


Source: L. Lazebnik

Corner Response Function

$$R = \det(M) - \alpha\, tr(M)^2 = \lambda_1\lambda_2 - \alpha(\lambda_1 + \lambda_2)^2$$

- $\det(M) = \lambda_1\lambda_2$ becomes large only when **both** eigenvalues are large
- $tr(M) = \lambda_1 + \lambda_2$ penalizes strong one-dimensional structure
- α controls how aggressively edges are rejected



Source: L. Lazebnik

Harris Corner Detector

1. Compute partial derivatives at each pixel
2. Compute second moment matrix *M* in a Gaussian window around each pixel

$$M = \begin{bmatrix} \sum_{x,y \in W} w(x,y) I_x^2 & \sum_{x,y \in W} w(x,y) I_x I_y \\ \sum_{x,y \in W} w(x,y) I_x I_y & \sum_{x,y \in W} w(x,y) I_y^2 \end{bmatrix}$$

C.Harris and M.Stephens, A Combined Corner and Edge Detector,
Proceedings of the 4th Alvey Vision Conference: pages 147—151, 1988.

Source: L. Lazebnik

Harris Corner Detector

1. Compute partial derivatives at each pixel
2. Compute second moment matrix *M* in a Gaussian window around each pixel
3. Compute corner response function R

C.Harris and M.Stephens, A Combined Corner and Edge Detector,
Proceedings of the 4th Alvey Vision Conference: pages 147—151, 1988.

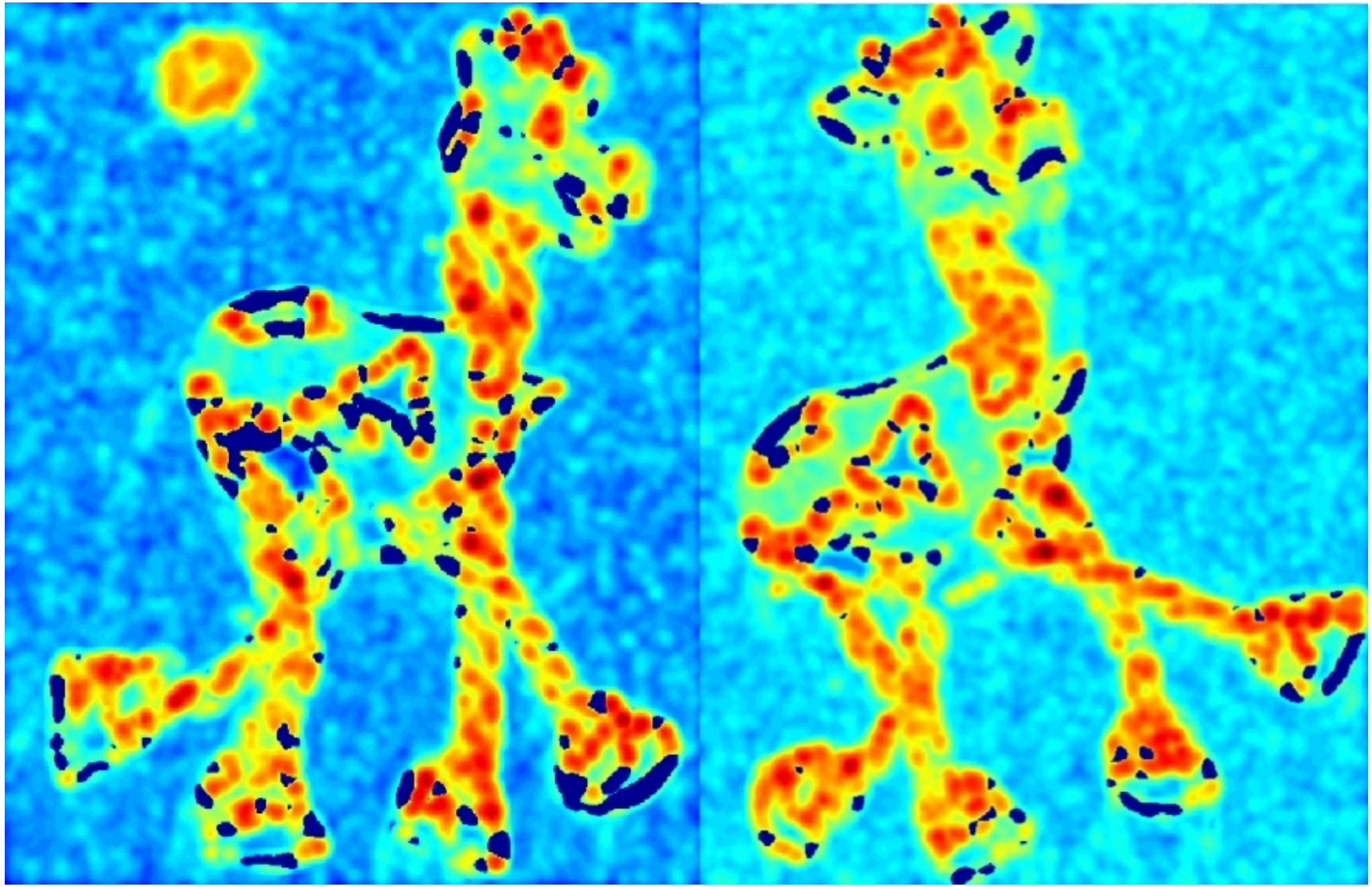
Source: L. Lazebnik

Example Image



Source: L. Lazebnik

Example Image



Compute corner response

Source: L. Lazebnik

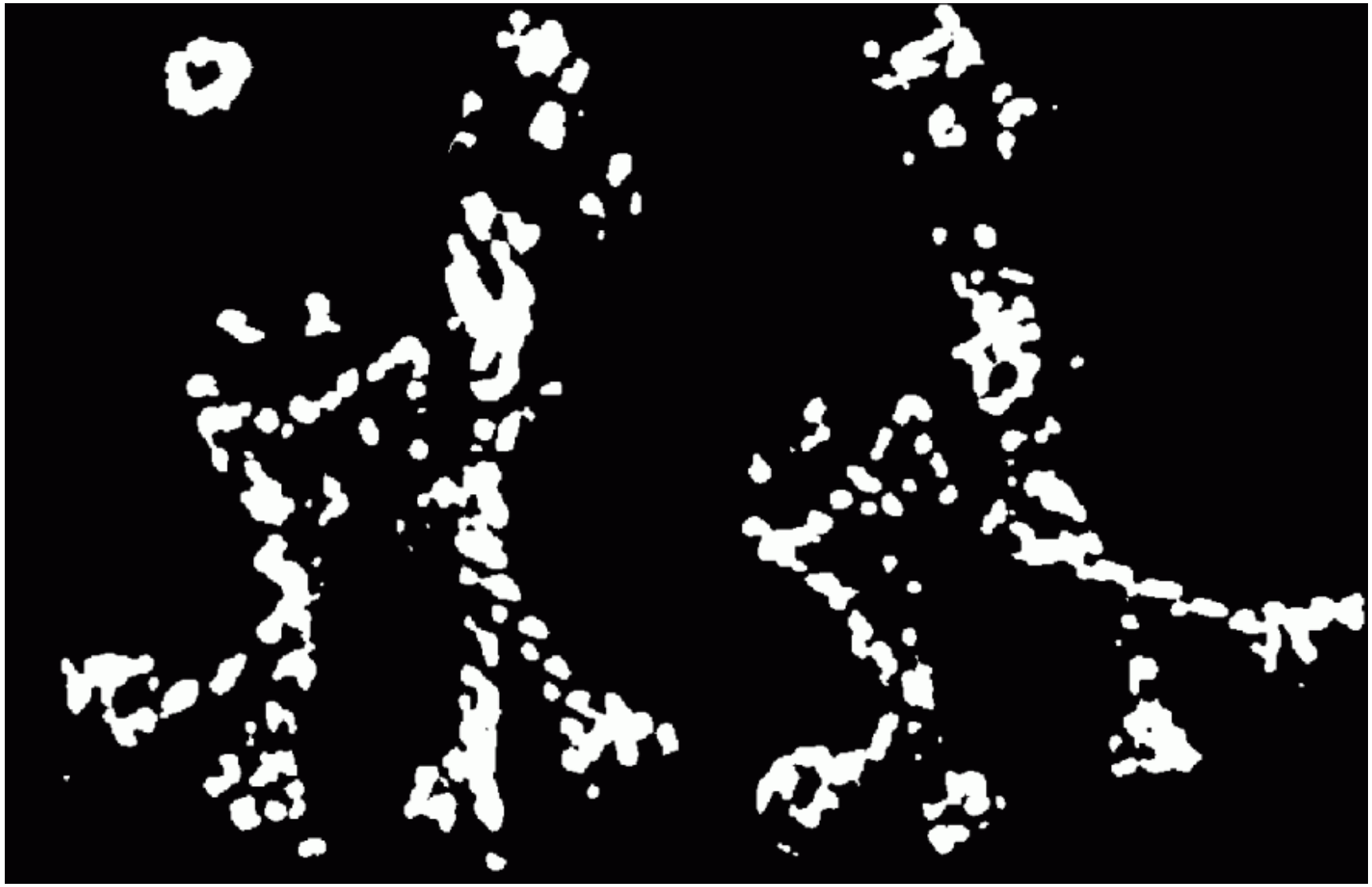
Harris Corner Detector

1. Compute partial derivatives at each pixel
2. Compute second moment matrix *M* in a Gaussian window around each pixel
3. Compute corner response function *R*
4. Threshold corner response function
5. Find local maxima of response function (NMS)

C.Harris and M.Stephens, A Combined Corner and Edge Detector,
Proceedings of the 4th Alvey Vision Conference: pages 147—151, 1988.

Source: L. Lazebnik

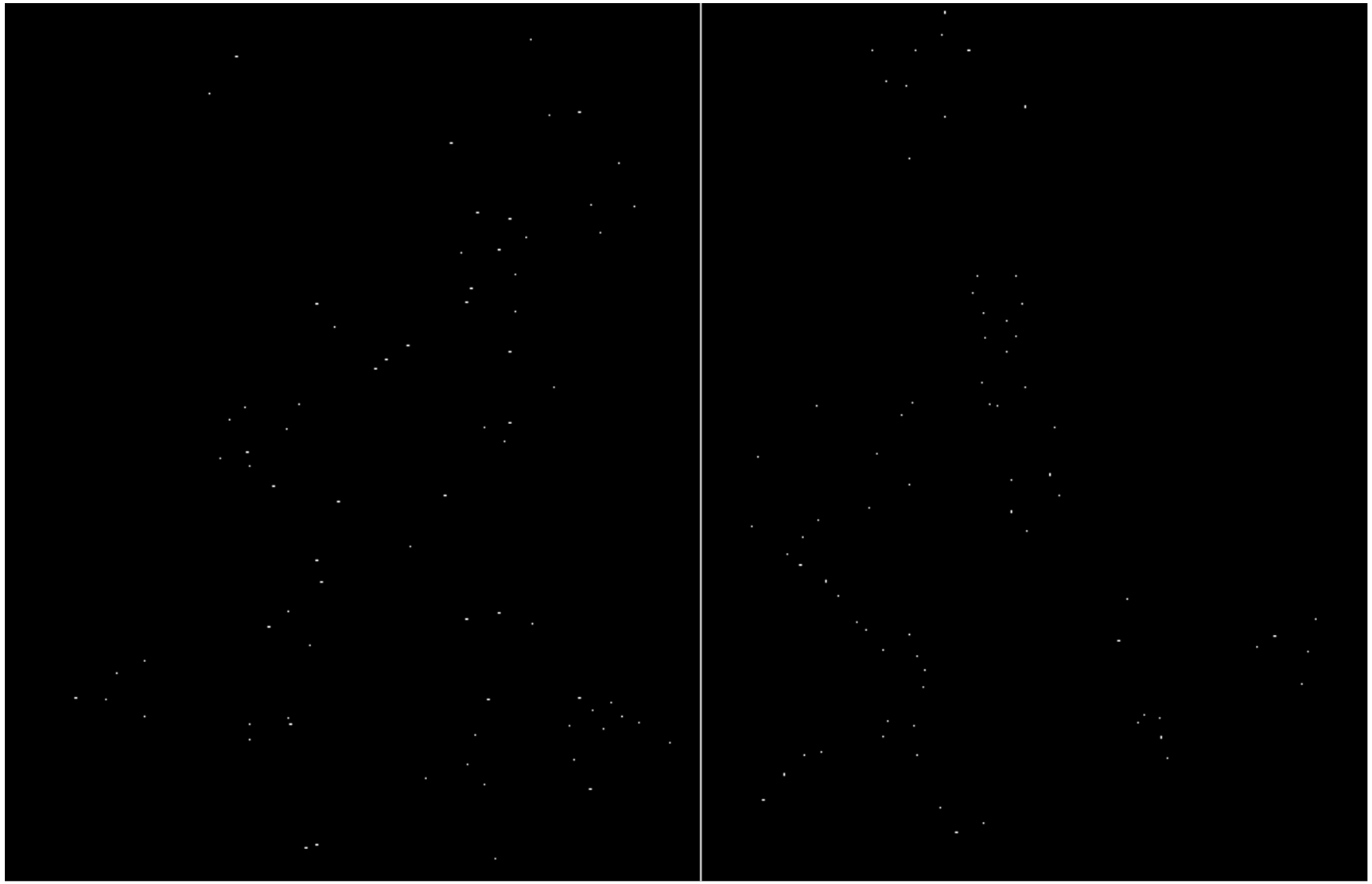
Example Image



Find points with large corner response: R > threshold

Source: L. Lazebnik

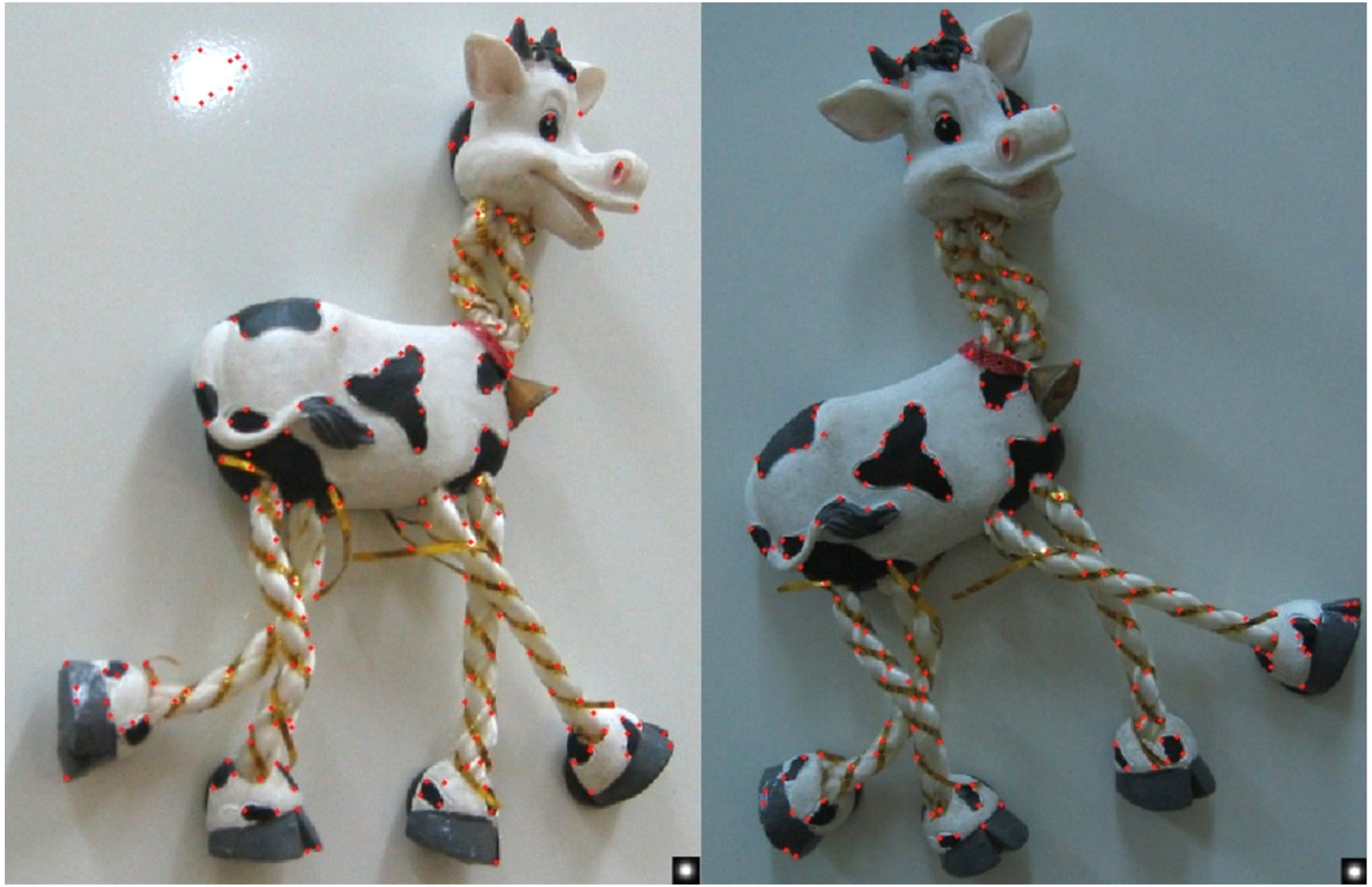
Example Image



Take only the points of local maxima of R

Source: L. Lazebnik

Example Image



Red points are detected corners

Source: L. Lazebnik

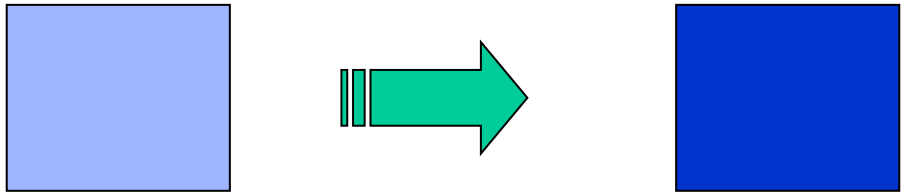
Robustness of Corner Features



What happens to corner features when the image undergoes geometric or photometric transformations?

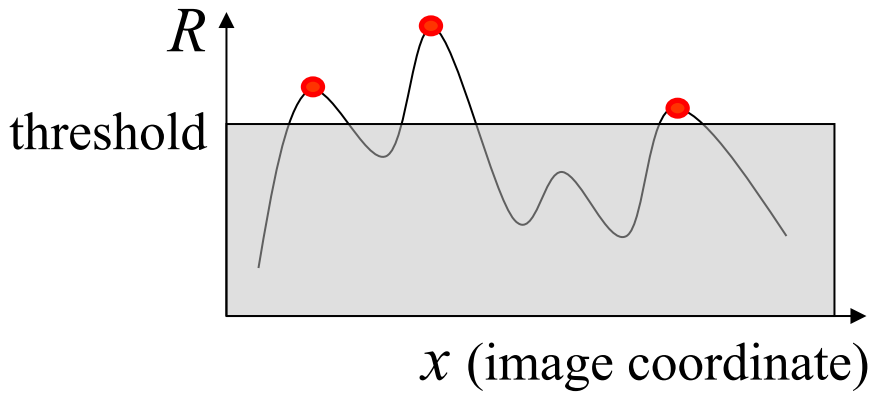
Source: L. Lazebnik

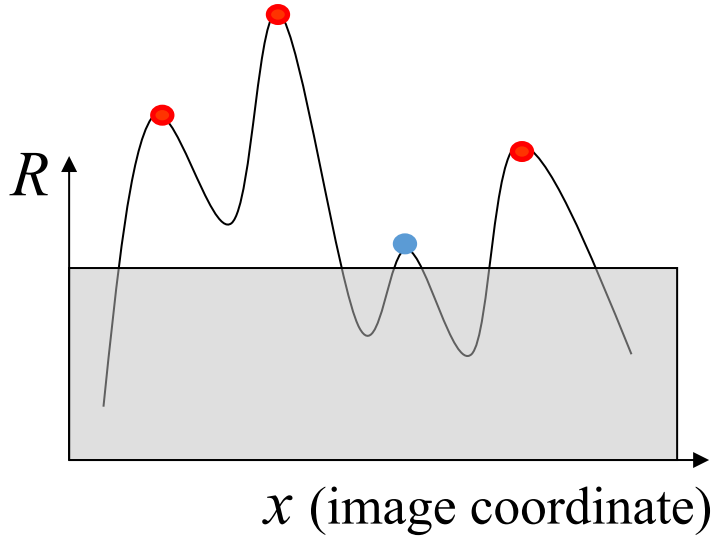
Affine Intensity Change



$$I \rightarrow aI + b$$

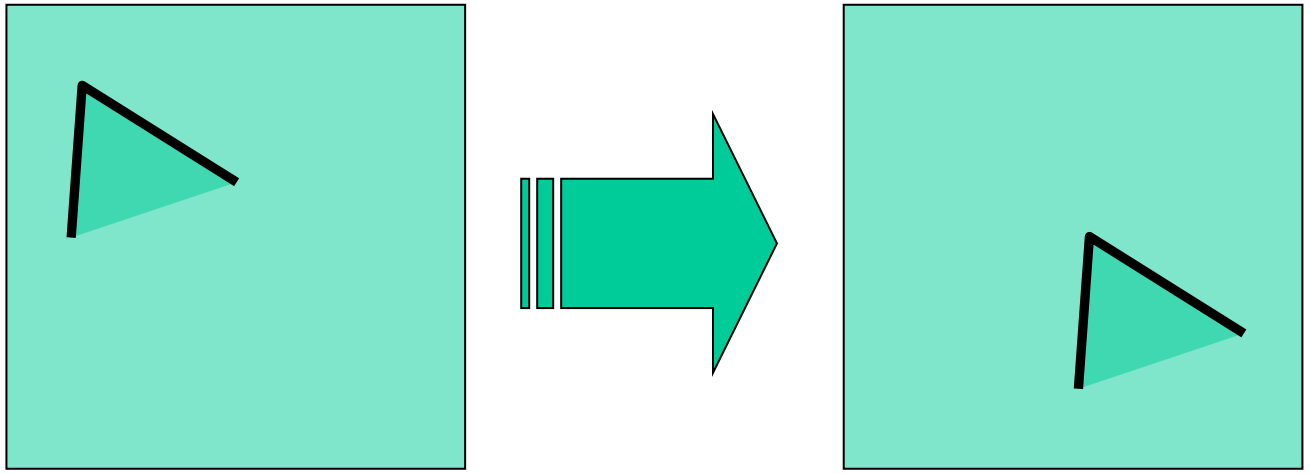
case	effect on corner features
If a=0, pure intensity shift:	invariant
If b=0, pure intensity scaling:	affecting thresholding





Source: L. Lazebnik

Image Translation

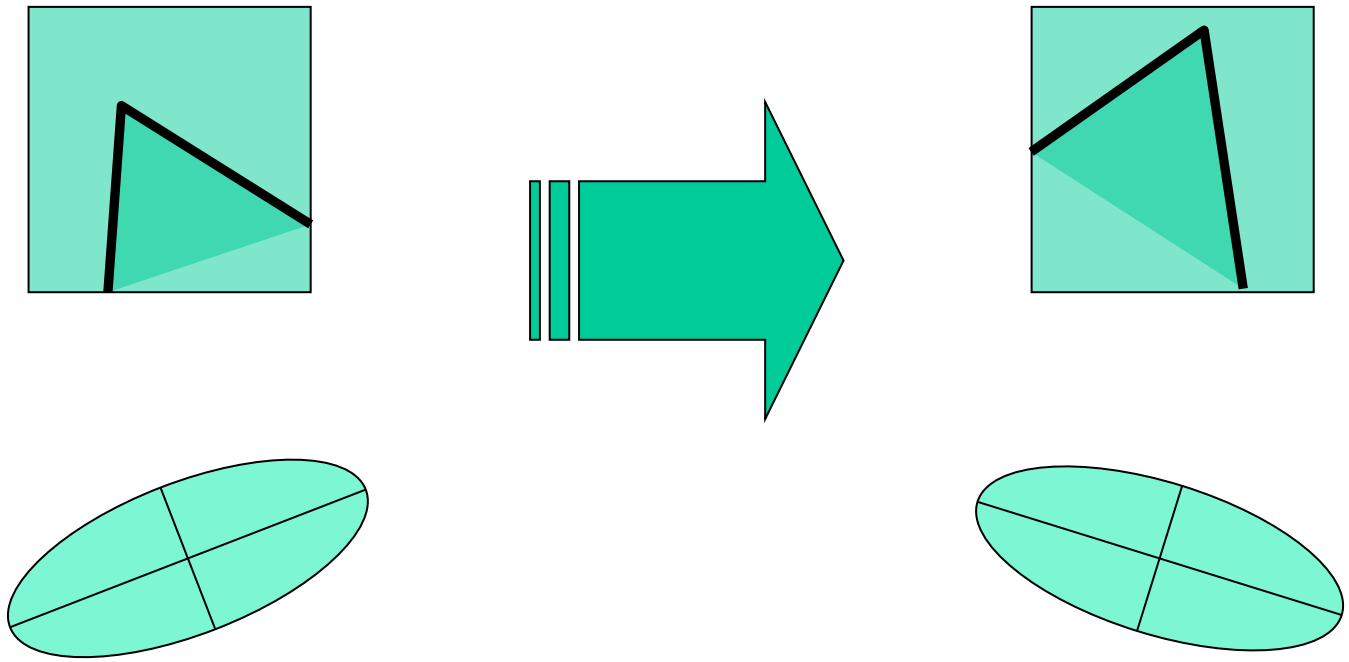


Derivatives and window function are shift-invariant

Corner location is covariant w.r.t. translation

Source: L. Lazebnik

Image Rotation

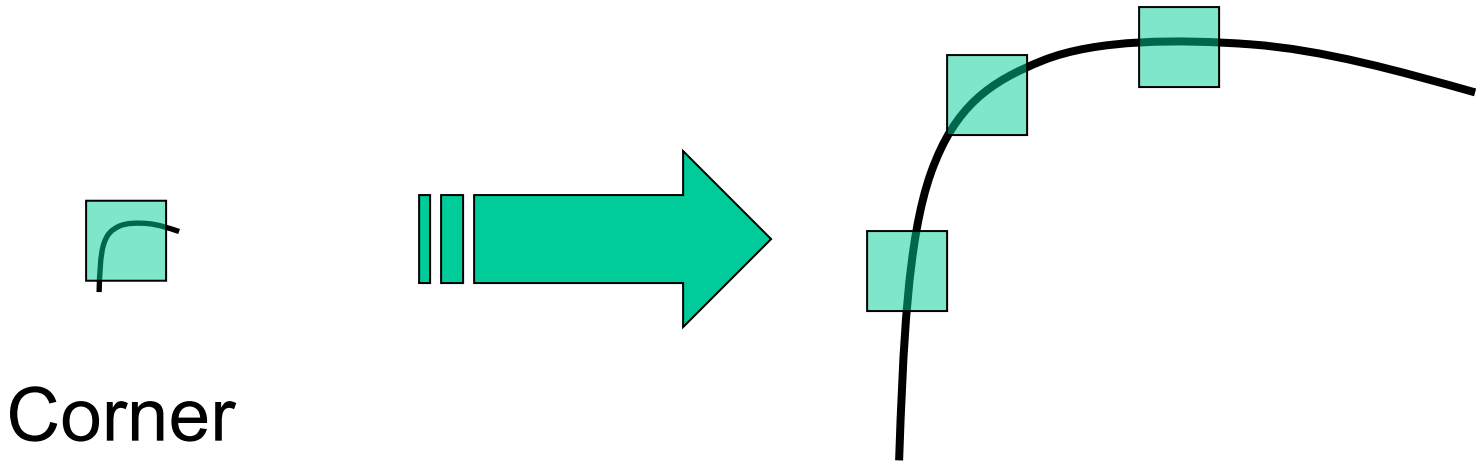


Second moment ellipse rotates but its shape remains the same

Corner location is covariant w.r.t. rotation

Source: L. Lazebnik

Scaling



Corner

All points will be classified as edges

Corner location is not covariant w.r.t. scaling

Source: L. Lazebnik

Summary

- Second moment matrix
- Harris detector: corner response function
- Harris is **translation/rotation covariant**, but **not inherently scale covariant**

Reading List

- **Szeliski, *Computer Vision: Algorithms and Applications*, 2e**
  - Ch.7.1–7.1.1: Points, Patches & Feature Detectors.
- **Forsyth & Ponce, *Computer Vision: A Modern Approach*, 2e**
  - Ch. 5.1 and 5.3–5.3.1: Image Gradients & Finding
- **Classic:** Harris & Stephens, *A Combined Corner and Edge Detector*, 1988.
- **Further:** Shi & Tomasi, *Good Features to Track*, CVPR 1994
- OpenCV, **Harris Corner Detection tutorial**